

S.L.K. SUTHAN

Email: suthan.95slk@gmail.com

Phone: +91-9003698836

Qualification:

Bachelor of Engineering (BE) in Mechanical Engineering from Easwari Engineering College, Chennai, India. **Year of Graduation:** 2016

Master of Technology (M.Tech) in Design Engineering from BITS Pilani (WILP). **Expected:** 2027

PROFESSIONAL SUMMARY

Functional Lead and Aerospace Engineer with 9+ years of experience in aero-engine systems design, flow control valves, performance modelling, and force margin analysis across Rolls-Royce and Parker Aerospace programs (UK, Germany, USA).

Extensive background in mechanical design and product definition, including pipes, brackets, ducts, sheet metal, and machined components, with expertise in GD&T(ASME Y14.5) and Model-Based Definition (MBD). Proven experience in valve design, spring selection and geometry optimization to achieve weight reduction and enhanced performance.

Specializes in physics-based simulation using MATLAB, Simulink, and Simscape, integrating fluid dynamics, actuator mechanics and performing force balance analysis using Free Body Diagrams (FBD).

PROFESSIONAL EXPERIENCE

Designation: IT Analyst– Tata Consultancy services, Bangalore

May 2026-Present

TCS-Parker Aerospace (May 2026 – Present)

Responsibility : Functional Lead – Performance Analysis

Technical Leadership

- Leading a team of 8 engineers in performance modeling and force margin analysis of aero-engine flow control valves.
- Defined standard engineering methodology, estimation guidelines, and modeling practices for performance analysis.
- Act as technical reviewer for models, calculations, and validation reports, ensuring engineering quality.

Performance Modelling & Simulation

- Developed high-fidelity models in MATLAB, Simulink, and Simscape to simulate valve performance
- Built multi-domain models integrating:
 - Fluid dynamics (mass flow, pressure drop)
 - Mechanical actuation systems
- Implemented governing physics:
 - Orifice-based flow modeling (compressible/incompressible)
 - Cv/Cd-based discharge relationships
 - Pressure-flow coupling across engine envelope
- Modeled nonlinear effects:
 - Stiction & Friction on Seal (Coulomb, viscous)
 - Leakage paths and internal clearances

Performance Modelling & Simulation

- Performed detailed component-level analysis:
 - Valve piston/poppet kinematics
 - Actuator response characteristics
 - Flow path behavior and restrictions
- Defined:
 - Boundary conditions (pressure, temperature, flow envelope)
 - Modeling assumptions (fluid properties, friction simplifications)
 - Solver configurations (stability and convergence control)
- Executed Structured workflow: Input Study – Model development – Debugging – Validation – Documentation

Force Margin Analysis

- Performed first-principles force balance calculations:
 - Pressure forces
 - Spring preload and stiffness effects
 - Seal Friction (static/dynamic)
 - Actuator forces
- Developed Free Body Diagrams (FBD) for valve components
- Evaluated force margin across full stroke and engine conditions
- Investigated worst-case scenarios:
 - Temperature extremes
 - Pressure limits
- Performed:
 - Stroke length validation
 - Cd-based flow-force interaction analysis

Verification and Validation

- Led technical reviews with cross-functional teams (Design, CFD, Testing)
- Co-related model predictions with test data and analytical results
- Developed
 - Model Documentation
 - Technical Reports
 - User guide and Validation summaries

TCS-Parker Aerospace (Jan 2024 to Aug 2025)

Design Leadership

- Led a Team of 10 engineers on 2D to 3D model conversion and product definition using SolidWorks.
- Ensure compliance with ASME standards and Parker Aerospace design guidelines.

Valve Design & Optimization

- Redesigned Anti-icing and flow control valves:
 - Achieved weight reduction and compact geometry
 - Improved packaging efficiency and performance
- Performed spring design calculations:
 - Stiffness selection
 - Preload optimization
 - Force alignment with operating requirements
- Supported integration of design with performance analysis models

Cross-Functional Collaboration

- Worked with:
 - Analysis teams (performance & stress)
 - CFD teams
 - Customer stakeholders

TCS-Rolls-Royce, UK and Germany (Dec 2016 – Dec 2023)

Responsibility – Product Definition Delegated checker | Onsite Co-ordinator | Design Engineer

Delegated Checker – RollsRoyce, Derby (UK)

- Verified and approved engine external pipes and bracket models/drawings with 100% compliance.
- Ensured adherence to:
 - GD&T standards (ASME Y14.5)
 - Fit, Form, Function and Manufacturability

Onsite Co-ordinator Experience – RollsRoyce, Germany

- Managed co-ordination between customer and offshore team (12 engineers) for the Fleet Modification Team.
- Reviewed and delivered Design Definition Report for the Controls Team.
- Designed and reviewed BR710 engine external components for the BR710 engine rebuild program.

Design Engineering

- Designed, Modelled and Product defined components using Unigraphics Nx:
 - Routed pipes
 - Sheet metal components
 - Machined brackets
 - Ducts and castings
- Applied:
 - GD&T (tolerancing and manufacturing control)
 - Model-Based Definition (PMI)
- Developed:
 - Core engine fairings
 - Complex fabrication and solution drawings
- Additional exposure:
 - P&ID systems
 - Harness form-board basics

ACADEMIC RECORD

- Course: B.E, Mechanical engineering
Name of the institution: Easwari engineering college, Chennai
Year of passing: 2016
CGPA: 8.6 CGPA
- Class: HSE, 12th
Name of the institution: D.V.D Higher Secondary School, Kanyakumari
Year of passing: 2012
Percentage: 92.8 %
- Class: S.S.L.C, 10th
Name of the institution: St.Tresa's Matriculation Higher Secondary School, Kanyakumari
Year of passing: 2010
Percentage: 88.6 %

KEY PROJECTS:

- Flow Control Valve Design & Performance Analysis - Parker Aerospace
- Anti-Icing Valve Optimization - Parker Aerospace
- Fleet Modification – Rolls Royce Germany
- Engine Rebuild Components – Rolls Royce Germany
- Engine Externals System – Rolls Royce Germany & UK

TECHNICAL SKILLS

Simulation and Analysis

MATLAB, Simulink, Simscape

CAD & Design

NX Unigraphics, CATIA V5, SolidWorks, AutoCAD

Engineering Expertise

- Physics-based Performance Modeling
- Force Margin Analysis
- Free Body Diagram (FBD)
- GD&T (ASME Y14.5)
- Model-Based Definition (MBD)

Domain Knowledge

Aero Engines, Flow Control Valves, Actuation Systems, Engine Externals

CERTIFICATIONS

- Mathworks – Simscape Fluid system and Physical system
- Agile Way of working experience
- Innovation & creativity foundation
- Team skills foundation

TRAININGS & LEADERSHIP

- Delivered training on GD&T and Model Based Definition (MBD)
- Mentored new engineers on Rolls Royce standards and Aerospace engineering practices.
- Established team level best practices in modelling and analysis workflows

ACHIEVEMENTS

- Certificate of appreciation from Chief of component engineering, Installations, Rolls-Royce plc for designing and delivering complex design brackets.
- Puzzle piece recognition award from Director of Business Aviation, Chairman, RR Germany for the DDR release (Pearl 15 program).
- Certificate of appreciation from Aerospace & Defense Unit Head TCS & Senior Engineer Manager, Parker Aerospace for Outstanding support on Selector Valve design Project.
- Star of the Decade 2019-2020, Tata consultancy services.
- Star of the month award, Tata consultancy services.
- 3 On the spot award, Tata consultancy services.
- Applause award, Tata consultancy services.
- Service and commitment award, Tata consultancy services.